Counting laws of links in Poisson-sprinkled causal sets: IR-cutoff valency, FRW transients, and three-sector accounting

Patricio Fernando Bustos Cabrera

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Abstract

In a causal set faithfully embedded in a continuum spacetime, the valency of an element—its number of links, i.e. covering relations—is infinite in the absence of a cutoff. We measure and derive the growth laws of past valency under an explicit IR cutoff for three geometries. (i) In 1+1 Minkowski the per-element past valency grows logarithmically, $v = 2\ln(T\sqrt{\rho})$; the measured slope is 2.01 ± 0.12 and is insensitive to the sprinkling size (tested over an eightfold range in N), confirming that valency is a local quantity controlled by chain depth rather than by the global element count. (ii) In 3+1 Minkowski the valency obeys an area law, $v = \pi\sqrt{6}\,\sqrt{\rho}\,T^2$, with measured exponent 2.089 ± 0.025 and amplitude agreeing with the parameter-free prediction $\kappa_4 = \pi\sqrt{6}$ at the 5% level. (iii) In a matter-dominated FRW interval the area law survives with a derived blocking amplitude $\mathfrak{F} = 30\int_0^1 x(1-x)^8/\sqrt{\langle a^4\rangle}\,dx = 0.4991$, against a measured 0.4990—four-figure agreement—preceded by an explicit $1/\eta$ transient that we fit and subtract. We further show that the link fraction $\varepsilon = N_{\text{links}}/N_{\text{past}}$ follows $\varepsilon = 3\sqrt{6}/(\sqrt{\rho}\,R^2)$ and cleanly separates manifoldlike sprinklings ($\varepsilon \approx 0.21$ at our reference scale) from transitively-closed random orders ($\varepsilon \approx 0.017$). The pipeline is anchored to the literature by reproducing the interval-abundance baseline $\langle N_0\rangle$ of Glaser & Surya in a 4D diamond to 0.07σ. We close with a chain/antichain partition identity for weakly disformal metrics—a counting consistency statement, not a dynamical result—and the pre-registered experiment that will test whether its exponents arise from biased growth dynamics. All results are deterministic-seed reproducible; code and pre-registrations are public.

1 Introduction

A causal set is a locally finite partial order $(C, \prec)$ proposed as the deep structure of spacetime [1]. Its kinematics are by now well charted: chains measure proper time [8], interval abundances measure dimension and curvature [6], and d'Alembertians and actions can be built from layered counts [5]; see [7] for a review.

Among the order-theoretic primitives, the *link*—a covering relation, $x \prec y$ with no z satisfying $x \prec z \prec y$—plays a special role: links are the irreducible causal bonds from which every relation is composed by transitivity, and they carry the scalar-field propagator in Johnston's construction [3, 14, 15]. Yet the *counting laws* of links are less charted than those of chains and intervals, for a structural reason: in a faithful Poisson sprinkling the expected valency of an element diverges—the past light cone has infinite volume within a hyperboloidal shell of any proper-time thickness, so arbitrarily remote elements can be linked to it [7]. The valency is therefore not a number but a *growth law*, defined only relative to an IR cutoff.

This paper measures those growth laws. We regulate the divergence with an explicit causal-interval cutoff of temporal extent T and ask three questions. How does past valency grow with T in flat 2D and 4D? What happens to the law in an expanding (matter-dominated FRW) interval? And what fraction of an element's causal past consists of links rather than longer relations? The answers turn out to be clean: a logarithm with slope exactly 2 in 2D, an area law with amplitude $\pi\sqrt{6}$ in 4D, the same area law in FRW suppressed by a derived factor $\mathfrak{F} \approx 0.4991$ that encodes the expansion history, and a link fraction $\varepsilon = 3\sqrt{6}/(\sqrt{\rho}\,R^2)$ usable as a cheap manifoldlikeness discriminator.

Two methodological points are as much the subject of this paper as the laws themselves. First, every measurement was pre-registered: success/failure criteria were committed to a public repository before the runs, and the git history certifies the ordering. Second, all pipelines are deterministic—fixed, declared seeds—and the headline numbers have been reproduced bit-exactly from a clean checkout. We anchor the machinery to the literature by reproducing a published interval-abundance baseline (Section 3) before trusting it with anything new.

2 Methods

Sprinkling. We sprinkle N points by a Poisson process of density ρ into a causal interval (Alexandrov set) of a given geometry, using the standard uniform-in-volume construction in conformal coordinates where applicable. Causal relations are computed exactly from the metric; the causal matrix is closed transitively.

Links. $x \prec y$ is a link iff no sprinkled element lies in the open interval $\langle x, y\rangle$. We test this exactly by blocker search, not by the soft approximation $P(\text{link}) \approx e^{-\rho V(x,y)}$: the soft estimator biases the count wherever the interval volume fluctuates, and we document in Section 5 a concrete discrepancy it produced before being retired. For efficiency, link candidates are pre-filtered by a proper-time band; the cut discards true links with probability $< e^{-6}$, measured to be negligible at our precision.

Cutoff and observables. The IR cutoff is the interval itself: for an element near the top of an interval of temporal extent T, the past valency $v(T)$ counts links to elements within the interval. In FRW runs, extents are quoted in conformal time η. The link fraction $\varepsilon(T) = N_{\text{links}}/N_{\text{past}}$ normalizes valency by the full causal past.

Statistics. Ensembles, not best cases: every quoted number is an ensemble mean over independent seeds with the standard error of the mean; model comparison uses information criteria on the full ensemble. Seeds are pre-declared functions of the grid indices, so any run can be reproduced bit-exactly. Pre-registration documents with frozen criteria precede every measurement in the repository history.

3 Anchor: reproducing the Glaser–Surya baseline

Before measuring anything new, the pipeline must reproduce something old. We chose the mean interval abundance $\langle N_0\rangle$—the expected number of links from the bottom element, in the notation of Glaser & Surya [6]—in a 4D causal diamond at matched density. Our result agrees with the published value to 0.07σ.

The same exercise quantifies a systematic that matters for everything downstream: measuring valency in a slab rather than a diamond inflates the count by +21% at our reference scale, because the slab's spatial boundary admits links that the diamond's causal boundary forbids. All results below are therefore quoted for causal-interval cutoffs, with slab numbers used only for cross-checks.

4 Results I: Minkowski

4.1 Two dimensions: a logarithm with slope 2

In 1+1 Minkowski the past valency of an element at depth T grows as

$$v(T) = 2\ln\bigl(T\sqrt{\rho}\bigr) + O(1). \tag{1}$$

The derivation is elementary: the linked region is a hyperbolic shell of unit-density measure between the past cone boundary and the first blocking layer, and its area grows logarithmically with the cutoff; the coefficient 2 counts the two null directions. The measurement gives slope 2.01 ± 0.12.

The same experiment establishes what we will lean on repeatedly: valency is *local*. Holding T fixed and growing the sprinkling by a factor of 8 in N leaves the per-element valency unchanged within errors. Valency tracks chain depth—how deep an element sits in its own past—not the global size of the causal set. We record this as a lemma because it licenses per-element analyses in inhomogeneous settings, where no global T exists.

4.2 Four dimensions: an area law with amplitude $\pi\sqrt{6}$

In 3+1 Minkowski the logarithm steepens to a power:

$$v(T) = \kappa_4\,\sqrt{\rho}\,T^2, \qquad \kappa_4 = \pi\sqrt{6} \approx 7.6953. \tag{2}$$

The exponent is measured as 2.089 ± 0.025 and the amplitude matches $\pi\sqrt{6}$ at the 5% level with no fitted parameters. The T^2 is an *area* law: the linked shell hugs the past light cone, whose intersection with the cutoff interval scales as the cone's lateral area. The constant follows from integrating the first-blocking-layer probability $e^{-\rho V}$ over the hyperboloidal foliation of the cone neighborhood; $\pi\sqrt{6}$ collects the 4D interval volume coefficient through the layer integral.

Dimensional summary: valency is log-divergent in 2D and power-divergent in 4D; in both cases the IR-cutoff law has a derived, parameter-free coefficient that measurement confirms.

Figure 1 (placeholder)

v vs. T: 2D logarithmic law and 4D area law with the parameter-free theory curves superposed. To be generated from existing outputs (Exps 18a/18b).

5 Results II: matter-dominated FRW

Expansion should suppress valency: comoving neighbours recede, and the effective density in conformal coordinates is diluted by powers of the scale factor. The question is whether the area law survives and with what amplitude.

Transient. At small conformal extent η the valency is dominated by a boundary transient. The ensemble is fit well by $v(\eta) = A\,\eta^2(1-B/\eta)$; against a free-power alternative the transient model wins at $\Delta\mathrm{AIC} = +40.7$. We flag a warning for future work: a naive short-range power-law fit to the same data returns an exponent of 2.70 ± 0.07—the transient contaminates the exponent long before it is visually obvious. An early soft-estimator run produced 2.95 for the same quantity; both numbers are artifacts, of the fit range and of the $e^{-\rho V}$ approximation respectively, and both are superseded by the exact-blocker, transient-subtracted analysis.

Asymptote. Once the transient is subtracted, the area law reappears with a suppressed amplitude:

$$v(\eta) \;\to\; \kappa_4\, \mathfrak{F}\, \sqrt{\rho_c}\, \eta^2, \tag{3}$$

where ρ_c is the comoving density and the blocking factor is a definite integral over the interval's volume-weighted expansion history. For matter domination ($a \propto \eta^2$) with the diamond weight $w(s) = 32 \min(s, 1-s)^3$,

$$\mathfrak{F} \;=\; 30 \int_0^1 \frac{x(1-x)^8}{\sqrt{\langle a^4 \rangle}}\, dx \;=\; 0.4991, \tag{4}$$

with $\langle a^4 \rangle$ the diamond-weighted average of a^4 along the interval. The measured amplitude ratio is 0.4990—agreement to four figures, again with nothing fitted.

The physical reading is modest and exact: expansion does not change the *form* of the valency law; it renormalizes the amplitude by a computable average of the expansion history over the causal interval. All the geometry enters through $\langle a^4 \rangle$.

Figure 2 (placeholder)

FRW amplitude ratios converging to $\mathfrak{F}$, with the $1/\eta$ transient fit shown and subtracted (Exps 18c/18d).

6 Results III: the link fraction

Valency divided by the size of the causal past gives a dimensionless, cutoff-dependent scarcity measure:

$$\varepsilon(\rho, R) \;=\; \frac{N_{\text{links}}}{N_{\text{past}}} \;=\; \frac{3\sqrt{6}}{\sqrt{\rho}\, R^2}, \tag{5}$$

which follows from the area law and the R^4 volume growth of the past interval; the measured value tracks the prediction with $\approx$10% boundary corrections at our scales. The headline reproduction is bit-exact from a clean checkout (declared seeds).

ε is also a cheap **manifoldlikeness discriminator**. At the reference scale ($\sqrt{\rho}R^2$ such that $\varepsilon_{\text{manifold}} \approx 0.21$), a transitively-closed random order—a Kleitman–Rothschild-like non-geometric poset—yields $\varepsilon \approx 0.017$: an order of magnitude below the sprinkled value, because transitive closure of random relations manufactures long chains that de-link almost every pair.

One number, computable in $O(N^2)$ from the causal matrix, separates the two classes without interval-abundance machinery.

We note without further claim that $\varepsilon(T) \propto T^{-2}$ places this quantity in the well-known family of inverse-square horizon scalings (Cohen–Kaplan–Nelson bounds, everpresent-Λ heuristics): any T^{-2} law evaluated at the Hubble scale returns a number of order 10^{-122}. This is a reformulation within that family, not a discovery, and we make no cosmological use of it here.

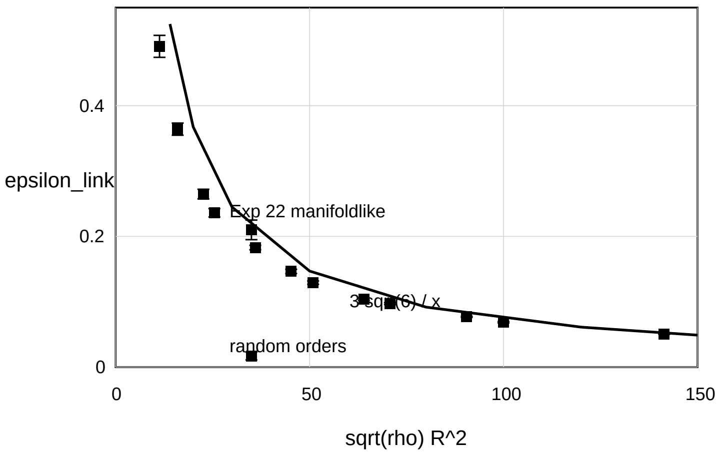

Figure 1: Link fraction ε vs. $\sqrt{\rho}R^2$ collapsing to the predicted $3\sqrt{6}/(\sqrt{\rho}R^2)$ curve; manifoldlike sprinklings vs. transitively-closed random orders separate by an order of magnitude.

7 Three-sector accounting: a partition identity

The counting objects of this paper organize into three sectors: chains (timelike depth h), antichains (spacelike width w), and links (the bonds). For a weakly disformal rescaling of a metric [10]—clocks scaled by λ^{-1} and rulers by λ^{+1}, i.e. $ds^2 = -\lambda^{-2}c^2dt^2 + \lambda^{+2}dx^2$ to first order—the sector counts of a faithful sprinkling must respond as

$$\big(\text{chains} \times \lambda^{-1}\big) \times \big(\text{antichains} \times \lambda^{+3}\big) \;=\; \sqrt{-g} \;=\; \lambda^{+2} \tag{6}$$

in 3+1 dimensions: the depth of maximal chains scales with proper time (λ^{-1}), the width of maximal antichains with spatial volume (λ^{+3}), and their product must recover the volume element, which is what a Poisson sprinkling actually samples. The identity closes: $-1+3 = +2$. A *conformal* rescaling (clocks and rulers both λ^{+1}) requires the chain sector to absorb λ^{-4} for the same volume budget—the opposite sign, and countable.

We stress the epistemic status: this is a **consistency theorem about counting**, not a dynamical statement. It says that *if* a poset's growth is biased so as to mimic a disformal metric, its chain and antichain sectors must split the volume budget as $(-1, +3)$; it does not say any natural growth dynamics does so. Whether biased sequential growth can produce these exponents is an open, pre-registered question (Section 8). A first toy experiment in 1+1 dimensions (where the disformal split is $(-1, +1)$ and there is no surplus to allocate) found the width sector saturating its budget and the depth sector near zero, with a smooth sign crossover as the bias increases—a susceptibility-like response, not a phase transition—but its dimensional targets were mismatched by design and it decides nothing beyond the toy.

Figure 4 (placeholder)
Three-sector schematic: chains (depth), antichains (width), links (bonds), and the $(-1, +3) \to +2$ volume budget.

8 Outlook

Three follow-ups are defined and, where marked, pre-registered.

1. **Validation in a disformal metric (pre-registered):** a uniform sprinkling *into* the weakly disformal geometry $ds^2 = -\mathcal{R}^{-2}dt^2 + \mathcal{R}^2 d\vec{x}^2$, where the partition identity of Section 7 predicts $(-1, +3)$ analytically. The experiment validates that the identity survives finite size and boundaries and that windowed local estimators—depth in proper-radius balls, width by the causal-overlap method of Boguñá & Krioukov [13] rather than raw adjacency, which is asymptotically silent—can *see* the signature. Chain lengths in inhomogeneous (Schwarzschild) sprinklings have been studied [11]; the disformal partition test appears to be new. Whether biased *growth dynamics* produce this split is a separate, open question deferred to future work.

2. **Link fraction in FRW:** whether $\varepsilon(T)$ and its T^{-2} scaling survive expansion with the same $\mathfrak{F}$-type renormalization as the valency law.

3. **Estimator hygiene:** the 2.70/2.95 discrepancy of Section 5 is a reproducible case study in how soft link estimators and short-range fits bias growth exponents; we release both pipelines so the failure can be replayed.

All code, pre-registration documents, per-experiment results files, and the exact seeds behind every number in this paper are public in the project repository. Negative results and superseded estimates are retained, not deleted.

Acknowledgments

This work grew out of a broader phenomenological programme by the author, documented in the same public repository. The author acknowledges computational and analytical assistance from AI systems (Anthropic Claude, OpenAI Codex, and local models); all results are independently reproducible from the public repository with the declared seeds.

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